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COMMENTARY

Vital Directions For Health And Health Care: Priorities For 2025

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ABSTRACT In 2016, ahead of the US presidential election, the National Academy of Medicine launched the strategic initiative Vital Directions for Health and Health Care—a series of papers on critical areas of US health care written by the nation's experts and intended to provide nonpartisan guidance to the incoming administration. The National Academy of Medicine continued the initiative in 2021. The current series, titled Vital Directions for Health and Health Care: Priorities for 2025, contains six articles on priority areas in US health and medicine that demand urgent attention. Here we provide an overview of the articles, which spotlight key areas for action and transformative change: revitalizing the biomedical research enterprise, modernizing public health, charting new directions for women's health, safely integrating artificial intelligence throughout the health care system, addressing the impact of climate change on health and equity, and transforming the US health care system to one that performs at par with the systems of other industrialized nations. Common themes include coordinating a strategic vision, correcting adverse economic incentives, investing in the health care workforce, reducing fragmentation, and emphasizing equity. Each article offers actionable priorities for the new administration to improve health outcomes for all Americans.

In 2016, under its congressional mandate as an adviser to the nation, the National Academy of Medicine launched a strategic initiative to gather expert insights on critical health and health care priorities. The resulting report, *Vital Directions for Health and Health Care*,[1] was published in 2017, providing the incoming US presidential administration, policy makers, opinion leaders, and the public with trusted, nonpartisan, evidence-based analysis of the most pressing opportunities and challenges in health, health care, and biomedical science. The initial Vital Directions series analyzed policy opportunities across nineteen critical areas to drive progress in three domains: better health and well-being, high-value health care, and robust science and technology. Four years later, in 2021, the second Vital Directions series devoted particular attention to compelling national challenges and opportunities in five areas: infectious disease threats, women's and children's health, health for older adults, mental health and addiction services, and health care costs and financing.[2] The assessment also addressed the new and continuing impacts of the COVID-19 pandemic.

With the arrival of 2025 and the inauguration of a new presidential administration, this third Vital Directions series highlights the critical issues that take on particular prominence, requiring leadership and decision making at multiple levels, especially from the White House and cab-

inet departments with key health leadership responsibilities. Since the publication of the 2021 series, the health, health care, and biomedical research landscape in the United States has undergone significant changes—in vulnerability awareness, the emergence of technologies, social and cultural circumstances, the health and health care consequences of investment patterns, and even the health implications of legislative and legal developments.

The experience of the COVID-19 public health emergency underscored systemwide vulnerabilities while also exposing new opportunities and shortfalls within the nation's public health and health care systems.[3] Emerging diseases with pandemic potential have surfaced, such as avian flu and mpox, with their full impacts yet to unfold. Persistent societal threats endure, including deepening health disparities between population groups and eroding levels of trust in the motivations and performance of various institutions. Although some of the loss of trust is due to the implications of private capital health care investments, especially worrisome is a new level of public distrust in science.

Advances in technology are having a transformative impact on health and science. Recent developments in artificial intelligence (AI), including the emergence of large language models and generative AI, will accelerate the generation and application of new insights for patient care and population health strategies. However, the trajectory of this technology is uncertain. AI may level access to health information and improve the ability to forecast the health and economic consequences of health products and services, yet without appropriate cautions, it could deepen disparities in health care and foster unintended consequences.

In addition, recent judicial actions are reshaping health activities at every level, leaving policy makers to face new complexities. The United States Supreme Court's June 2024 overturning of the 1984 *Chevron U.S.A., Inc. v. Natural Resources Defense Council, Inc.* ruling may significantly limit federal agencies' ability to interpret laws and take enforcement actions. In addition, the June 2022 *Dobbs v. Jackson Women's Health Organization* decision overturning *Roe v. Wade* holds implications for health and health care delivery organizations throughout the country.

Health Priorities For 2025

Six articles published in this issue of *Health Affairs* build on the original Vital Directions series by spotlighting critical areas that demand immediate leadership and decisive action. The articles focus on revitalizing the biomedical research enterprise, modernizing public health, improving women's health, safely integrating AI throughout the health care system, addressing the impacts of climate change on health and equity, and undertaking transformative changes to improve health care system performance in the US. Each article assesses the current landscape, identifies key priorities, and offers actionable strategies for progress. Common themes across the series include the need for coordinated, strategic leadership; reducing fragmentation in the health care system; correcting adverse economic incentives; investing in a robust health care workforce; and prioritizing equity to achieve the best possible health outcomes for all Americans. As a new administration takes office, this third set of articles in the Vital Directions series provides critical review and guidance for some of the most important decisions currently facing leaders at the national, state, and local levels in the United States, not to mention the world.

REVITALIZING THE US BIOMEDICAL RESEARCH ENTERPRISE Although the US biomedical research enterprise boasts a legacy of groundbreaking discoveries and innovations, it now faces significant challenges that jeopardize its global leadership and ability to serve all Americans effectively. Persistent poor population health outcomes, driven by rising chronic diseases and the widening gap between healthy life expectancy and actual lifespan, are key concerns. Meanwhile, federal investment in biomedical research has stagnated, and the fragmented funding landscape leaves the field without a unified vision. The workforce pipeline is also strained. Career pathways are unclear, and a lack of workforce diversity stifles both innovation and equitable health outcomes.

In their article, "Four Opportunities to Revitalize the US Biomedical Research Enterprise," E. Albert Reece and colleagues propose a comprehensive overhaul of the biomedical research landscape. At the heart of this vision is the creation of an independent, multidisciplinary advisory body to develop a strategic plan aligning research efforts with national health priorities and coordinating efforts across sectors. The proposal highlights the need to nurture and diversify the biomedical workforce, with a focus on strengthening education programs at minority-serving institutions and creating clearer career pathways for young researchers, modeled after successful initiatives such as the GI Bill. This reimagined pipeline would foster a more inclusive environment, welcoming international talent and investing in social sciences while prioritizing health equity in research. The article emphasizes addressing the upstream causes of health disparities, improving community-based

care, and enhancing science communication to build public trust. Recognizing the need for innovative funding, it advocates for new models that balance public benefit with market forces, encouraging partnerships between the private, public, and philanthropic sectors and supporting high-risk, high-reward research to improve population health outcomes.

By fostering a more coordinated, diverse, and equitable research ecosystem, the authors aim to enhance health outcomes, promote equity, and extend healthy longevity for all Americans while preserving and enhancing the nation's global leadership in biomedical innovation. This vision is not merely a set of reforms, but a bold reimagining of how biomedical research can become a cornerstone of national health and prosperity.

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MODERNIZING PUBLIC HEALTH Despite advancements in medicine, the US public health system faces urgent challenges in its ability to protect and promote the health of all citizens. Declining life expectancy, persistent health disparities, and emerging health risks highlight the need for a modernized approach. The COVID-19 pandemic further exposed deep vulnerabilities, including resource shortages, data limitations, and fragmented funding. In addition, the field lacks support for addressing public health risks during nonemergency times, suffers from unclear roles within the broader health system, and faces diminishing public trust in its recommendations.

In "Updating US Public Health for Healthier Communities," Mark McClellan and colleagues propose a future in which public health operates as a "team sport," fostering partnerships across health care, community organizations, and the private sector. The authors call for federal support and administrative action to build an integrated infrastructure that unites public health, prevention-focused health care, and community-based organizations. If shared population health and health equity goals are advanced, the authors argue, existing funds can be used more effectively, with clearer guidance on how additional funding and authority can further improve outcomes. This collaborative vision seeks to create a more responsive, efficient, and community-driven public health system.

The authors outline four key areas for transformation: enabling cross-sector collaboration, aligning financing mechanisms with accountability for population health outcomes, improving data systems, and building a ready workforce. They advocate moving away from fragmented, short-term funding and seeking stable, outcome-oriented financing models, including harmonized quality measures across federal agencies and linking funding to improved population health outcomes. The authors also emphasize the need for data modernization to provide real-time insights, enabled by standardized data sharing between public health and health care partners. The authors conclude that addressing workforce shortages through cross-functional leadership programs, AI training, and stronger community partnerships would better equip public health professionals to respond to crises and tackle the chronic conditions that impede community well-being.

CHARTING NEW DIRECTIONS FOR WOMEN'S HEALTH The US consistently underinvests in and underprioritizes women's health. Women spend disproportionately more time in poor health compared with men,[4] and women and their families suffer from alarmingly high maternal mortality rates.[5] These persistent challenges adversely affect women's well-being across the lifespan, and they are compounded by stark disparities across demographic groups and a narrow definition of what constitutes women's health. Historically, the field has focused on reproductive health, neglecting the broader array of health issues women face throughout their lives. This limited perspective is exacerbated by a lack of research that accounts for both biological sex and gender, as well as insufficient diversity in clinical trials. As a result, the health care system often fails to meet the complex health needs of women across various ages, races and ethnicities, and socioeconomic backgrounds. The health care workforce mirrors these systemic issues. Although women make up the majority of health care workers, they face substantial barriers to career advancement, leadership, and equitable pay. This gender disparity affects not only those working within the system

but also the quality of care provided to female patients, as research shows that gender concordance between patients and providers can affect health outcomes.

To address these deep-rooted issues, Paula Johnson and colleagues call for a paradigm shift in their article, "New Directions for Women's Health: Expanding Understanding, Improving Research, Addressing Workforce Limitations." They propose a holistic, lifespan approach to women's health that extends beyond traditional boundaries. This new vision calls for broadening the definition of women's health, reimagining the research ecosystem to be more inclusive and representative, and addressing workforce limitations that hinder progress. The authors call for the creation of an interagency task force to develop a comprehensive, governmentwide approach to women's health, focused on translating research into tangible improvements in health outcomes. If data collection and reporting on sex, gender, and other demographic factors in federally funded research are improved, a clearer understanding of women's health can emerge, guiding more-targeted interventions.

Women's health is not a niche issue, but a fundamental pillar of public health and societal well-being. In 2025 and beyond, the opportunity to reshape women's health care in the US promises a future where all women can live longer, healthier lives.

PROMOTING SAFE, EFFECTIVE, AND TRUSTWORTHY AI USE The use of AI holds immense potential to revolutionize patient care and health outcomes, with this promise likely to materialize much sooner than many anticipate. This potential compels particular vigilance to ensure the safe, effective, and trustworthy implementation of AI. The health care sector continues to face issues such as unclear governance and standards for AI systems, workforce gaps in AI competency, the need for targeted research investments, and uncertainty around liability in AI-assisted decision making.

In their article, "Artificial Intelligence in Health and Health Care: Priorities for Action," Michael Matheny and colleagues propose addressing these challenges through four key areas. First, they call attention to the need to ensure the safe, effective, and trustworthy use of AI. This can be achieved through the development of clear definitions and standards; incentives for equitable AI adoption; standardized implementation processes; and dynamic, learning-based regulatory approaches. Continuous monitoring and mitigation of bias must be central to these efforts.

Second, building an AI-competent health care workforce is essential, as many providers lack familiarity with emerging technologies. This involves integrating AI education into health care curricula, expanding training to health-adjacent fields, and clearly defining the composition of an AI-competent workforce. These initiatives aim to equip health care professionals at all levels to effectively use AI in their practice.

Third, advancing AI research is critical to progress. This includes deepening scientific understanding of disease mechanisms, expanding AI applications in drug discovery and precision medicine, improving AI integration into clinical workflows, and investing in health care operations research. Addressing fairness, ethics, data quality, privacy, and explainability in AI systems will be paramount as technologies evolve.

Last, clarifying responsibility and liability in AI use is essential. The authors call for model licensing and indemnification terms to create a clearer legal framework for AI in health care.

Realizing AI's potential in health care requires a careful balance between innovation and safety, supported by policies that promote equitable access and adoption. Achieving this will require coordinated efforts from government agencies, health care organizations, educational institutions, and industry partners. When the inherent challenges are addressed, AI can be harnessed to significantly improve health outcomes and health care delivery. With careful stewarding, AI can usher in a new era of accelerated health data use and personalized health care for all Americans.

CRITICAL STEPS TO ADDRESS CLIMATE, HEALTH, AND EQUITY Climate change presents an unprecedented threat to human health and well-being in the United States and worldwide. More frequent and extreme weather events directly affect human health and well-being, with staggering economic and societal costs. Beyond acute events, chronic impacts include shifting disease vectors, worsening air quality, and degraded water quality. These effects are already disproportionately affecting socially and economically disadvantaged communities, exacerbating existing health disparities.

To address these challenges, William Frist and colleagues outline a multifaceted approach focused on three key areas in their article, "Critical Steps to Address Climate, Health, and Equity." First, the authors call for a transition to clean energy sources by halting new fossil fuel infrastructure, repealing fossil fuel subsidies, and increasing funding for zero-carbon energy sources. These changes could result in long-term health benefits outweighing the required investments. For example, nationwide efforts to eliminate energy-related emissions across the electric, transportation, building, and industrial

sectors could prevent 53,000 premature deaths per year and save $610 billion annually.[6]

Second, the US health care sector is responsible for approximately 8.5 percent of total US greenhouse gas emissions.[7] Although efforts are under way to reduce these emissions, challenges remain in financing and implementing mitigation upgrades, particularly for smaller facilities. Still, investment is worthwhile: Upgraded health care infrastructure can not only reduce emissions and improve air quality but also better withstand power outages, lack of fuel, and increases in patient load during disasters.

Last, with these coming changes, more resilience and capacity are needed within public health and health care to adequately respond to the consequences. This will involve improving climate-related health monitoring and enhancing public health preparedness for climate-related challenges. Many health care providers lack evidence-based guidance on how to recognize, prevent, and control medical conditions caused or exacerbated by climate change, such as cardiovascular or pulmonary events. New and existing evidence can be synthesized to develop evidence-based provider guidelines for the medical management of climate-sensitive conditions.

The authors also emphasize the need for continued research to strengthen the evidence base on the health impacts of climate change; guide effective interventions; and develop clear, impactful communication strategies to build public support and momentum for action. With concerted effort, the health sector can play a vital role in mitigating climate change and protecting population health.

IMPROVING HEALTH CARE SYSTEM PERFORMANCE Despite the wealth and resources of the United States, the health care system struggles to foster a healthy and thriving population. Life expectancy in the US trails six years behind that in leading nations, while chronic disease rates and medical errors are on the rise.[8–10] Persistent and worsening health disparities among people in certain racial, ethnic, and socioeconomic groups further exacerbate the problem.[11,12] Overall, the returns are abysmal for a nation that invests 17 percent of its GDP in health,[13] with health outcome rankings far down the list for nations spending half as much. Medical debt remains the leading cause of personal bankruptcy. Although certain areas of US health care excel globally, the system has long prioritized profit over patients' well-being, resulting in a convoluted structure that fails to meet the needs of all Americans. Decades of debate over market-based versus government-regulated solutions have produced a muddled middle ground, with no comprehensive approach being fully implemented or evaluated.

Understanding the most significant health challenges can help the new administration systematically advance multiple priority areas in health, medicine, and biomedical research.

In "From Laggard to Leader: Why Health Care in the United States Is Failing, and How to Fix It," Donald Berwick and colleagues propose a bold transformation of the US health care system. They envision a future where all Americans have access to affordable and high-quality care, racial and ethnic health disparities are eliminated, and health care spending is aligned with norms from peer countries. This transformation will require aligning payment incentives with population health goals, replacing fee-for-service payment, equalizing Medicaid and Medicare rates, strengthening primary care, and investing in health equity and social determinants of health. The authors also call for greater transparency in pricing and performance, the redesign of community benefit requirements for nonprofit hospitals, and the introduction of a public option for health insurance. A key element of these reforms is the establishment of stronger oversight and accountability mechanisms, including the creation of an independent organization with the authority to monitor progress and recommend policy changes.

Although implementing these reforms will be challenging, the urgent state of the nation's health and its financial impact demand swift and decisive action. With focused effort and political will, the United States can create a health care system that serves its people effectively, curbs excessive spending, and becomes a global model of excellence.

Building Momentum Across Sectors

Understanding the most significant health challenges facing Americans can help the new administration systematically advance multiple priority areas in health, medicine, and biomedical research. The articles in this issue outline a path forward to prepare the nation for ongoing and complex health challenges while building a future in which health is accessible to all Americans, regardless of race, ethnicity, age, geography, or socioeconomic status. By focusing on transforming biomedical research, modernizing public health, addressing women's health, integrating AI responsibly, taking action on climate change, and overhauling the health care system, these proposals provide actionable strategies to create a more equitable and resilient health care landscape. The National Academy of Medicine will continue to collaborate with partners across the nation to address these six critical priorities, eliminate health disparities, and improve the health and well-being of every American. ■

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NOTES

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